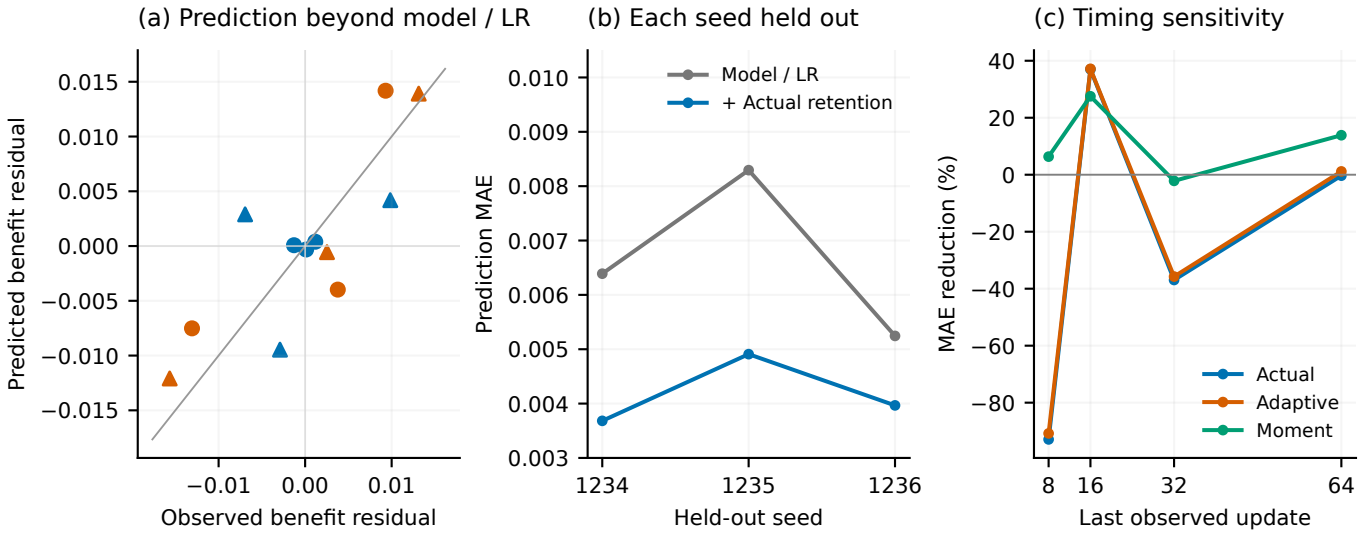

(a) Prediction beyond model / LR
Predicted benefit residual
Observed benefit residual
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.015
−0.01
0.00
0.01
(b) Each seed held out
Prediction MAE
Held-out seed
Model / LR
+ Actual retention
0.010
0.009
0.008
0.007
0.006
0.005
0.004
0.003
1234
1235
1236
(c) Timing sensitivity
MAE reduction (%)
Last observed update
40
20
0
−20
−40
−60
−80
8
16
32
64
Actual
Adaptive
Moment